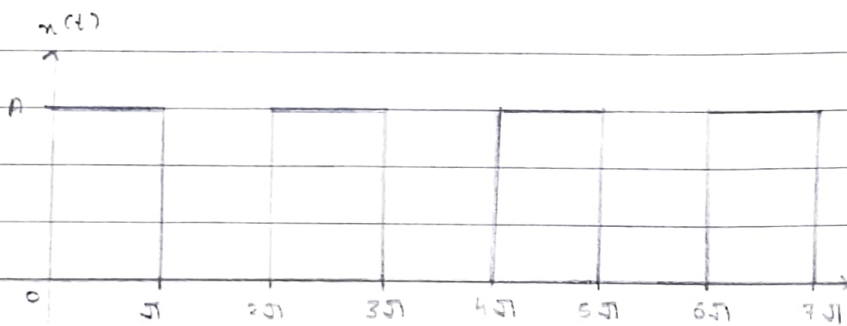


Assignment 3

Fourier Series

Q12



$$T = 2\pi$$

Time period of periodic wave form

$$\omega = 2\pi f = 2\pi \cdot \frac{1}{T} = 1$$

$\omega = \text{omega}$

$f = \text{frequency}$

$$f = 1/T$$

$$a_0 = \frac{1}{T} \int_0^T x(t) \cdot dt$$

$$a_n = \frac{2}{T} \int_0^T x(t) \cdot \cos(n\omega t) dt$$

$$b_n = \frac{2}{T} \int_0^T x(t) \cdot \sin(n\omega t) dt$$

$$a_0 = \frac{1}{T} \int_0^{2\pi} A dt$$

$$= \frac{A}{T} \int_0^{2\pi} 1 \cdot dt + \frac{1}{T} \int_{\pi}^{2\pi} 0 dt$$

$$= \frac{A}{T} [T]_0^{2\pi} + 0$$

$$= \frac{A}{T} [2\pi - 0] + 0$$

$$= \frac{2\pi A}{T}$$

$$= \frac{\pi A}{2\pi}$$

$$= \frac{A}{2}$$

$$a_n = \frac{2}{T} \int_0^T A \cdot \cos(n\omega t) dt$$

$$\frac{2A}{T} \left[\frac{\sin(n\omega t)}{n\omega} \right]_0^T$$

$$= \frac{2A}{Tn\omega} [\sin(n\pi) - \sin(n \cdot 0)]$$

$$= \frac{2A \sin(n\pi)}{2\pi n \omega}$$

$$= \frac{A}{\pi n \omega} \cdot 0 = 0$$

$$b_n = \frac{2}{T} \int_0^T A \sin(n\omega t) dt$$

$$= \frac{2A}{T} \int_0^T \sin(n\omega t) dt$$

$$= \frac{-2A}{Tn\omega} [\cos(n\pi) - \cos(0)]$$

$$= \frac{-A}{\pi n \omega} ((-1)^n - 1)$$

$$= \frac{A}{\pi n \omega} - \frac{A}{\pi n \omega} (-1)^n$$

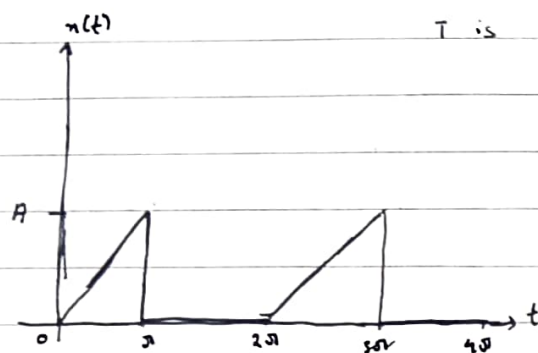
$$n = \text{even}, 0$$

$$= \frac{A}{\pi n \omega} - \frac{A}{\pi n \omega} = 0$$

$$n = \text{odd}, 1$$

$$= \frac{A}{\pi n} + \frac{A}{\pi n} = \frac{2A}{\pi n}$$

Find FS for



T is repeating term over time

$$x(t) \neq t$$

π is always small t

$$\sin(n\pi) = 0$$

$$\cos(n\pi) = (-1)^n$$

$$\sin(2n\pi) = 0$$

$$\cos(2n\pi) = 1$$

$$T = 2\pi$$

$$\omega = 2\pi f = 2\pi \cdot \frac{1}{T} = 2\pi \cdot \frac{1}{2\pi} = 1$$

$$n = \frac{A - 0}{\pi - 0} = \frac{A}{\pi}$$

$$x(t) = \begin{cases} A \left(\frac{t}{\pi} \right) & 0 \leq t \leq \pi \\ 0 & \pi \leq t \leq 2\pi \end{cases} \quad \text{y.m.n.c}$$

$$A_0 = \frac{1}{T} \left(\int_0^{\pi} \frac{At}{\pi} dt + \int_{\pi}^{2\pi} 0 dt \right)$$

$$= \frac{1}{2\pi} \cdot \frac{A}{\pi} \int_0^{\pi} t dt = \frac{A}{2\pi^2} \left[\frac{t^2}{2} \right]_0^{\pi}$$

$$= \frac{A \pi^2}{4 \pi^2} = \frac{A}{4}$$

$$A_n = \frac{2}{T} \int_0^{\pi} \frac{At}{\pi} \cos(n\omega t) dt + \int_{\pi}^{2\pi} 0 dt$$

$$= \frac{2}{2\pi} \cdot \frac{A}{\pi} \int_0^{\pi} t \cos(n \cdot 1 \cdot t) dt$$

$$= \frac{A}{\pi^2} \int_0^{\pi} t \cdot \cos(nt) dt$$

$$= \frac{A}{\pi^2} \left(t \cdot \frac{\sin(nt)}{n} - \int 1 \cdot \frac{\sin(nt)}{n} \right)$$

$$= \frac{A}{\pi^2} \left(\frac{t \sin(nt)}{n} - \left(-\frac{\cos(nt)}{n^2} \right) \right)$$

$$= \frac{A}{\pi^2} \left[\frac{t \sin(nt)}{n} + \frac{\cos(nt)}{n^2} \right]_0^{\pi}$$

$$= \frac{A^2}{\pi} \left[\left(\frac{\pi \sin(n\pi)}{n} + \frac{\cos(n\pi)}{n^2} \right) - \left(0 + \frac{\cos(0)}{n^2} \right) \right]$$

$$= \frac{A^2}{\pi} \left[\frac{\pi \sin(n\pi)}{n} + \frac{\cos(n\pi)}{n^2} - \frac{1}{n^2} \right]$$

$$= \frac{A^2}{\pi} \left\{ \frac{(-1)^n - 1}{n^2} \right\}$$

$$n=1$$

$$= \frac{A^2}{\pi} \left[\frac{-1-1}{1^2} \right]$$

$$= -\frac{2A^2}{\pi n^2}$$

$$n=2$$

$$\frac{A^2}{\pi} \left[\frac{(-1)^2 - 1}{4n^2} \right]$$

$$= 0$$

$$b_n = \frac{2}{T} \int_0^{\pi} \frac{At}{\pi} \sin(n\omega t) dt$$

$$= \frac{2}{2\pi} \cdot \frac{A}{\pi} \int_0^{\pi} t \sin(nt) dt$$

$$= \frac{A}{\pi^2} \int_0^{\pi} t \cdot \sin(nt) dt$$

$$= \frac{A}{\pi^2} \left(t \cdot \frac{-\cos(nt)}{n} - \int 1 \cdot \frac{-\cos(nt)}{n} \right)$$

$$= \frac{A}{\pi^2} \left(\frac{-t \cos(nt)}{n} + \frac{\sin(nt)}{n^2} \right)$$

$$= \frac{A}{\pi^2} \left(\frac{-\pi \cos(n\pi)}{n} + \frac{\sin(n\pi)}{n^2} - (0) \right)$$

$$= \frac{A}{\pi^2} \left[\frac{-\pi (-1)^n}{n} \right]$$

$$= \frac{-A (-1)^n}{\pi n}$$

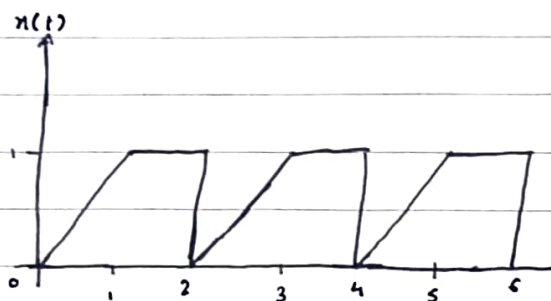
$$n=1$$

$$\frac{+A}{\pi n}$$

$$n=2$$

$$\frac{-A}{2\pi n}$$

Find FS for:



$$T = 2$$

$$\omega = 2\pi f = 2 \cdot \pi \cdot \frac{1}{2} = \pi$$

$$x(t) = \begin{cases} t & 0 \leq t \leq 1 \\ 1 & 1 \leq t \leq 2 \end{cases}$$

$$a_0 = \frac{1}{T} \left(\int_0^1 t \cdot dt + \int_1^2 1 \cdot dt \right)$$

$$= \frac{1}{2} \left[\left[\frac{t^2}{2} \right]_0^1 + \left[t \right]_1^2 \right]$$

$$= \frac{1}{2} \left[\frac{1}{2} + 1 \right]$$

$$= \frac{3}{4}$$

$$a_n = \frac{1}{T} \left[\int_0^1 t \cdot \cos(n\pi t) dt + \int_1^2 1 \cdot \cos(n\pi t) dt \right]$$

$$= \frac{1}{2} \left[\left[\frac{t \sin(n\pi t)}{n\pi} - \int \frac{\sin(n\pi t)}{n\pi} \right] + \frac{\sin(n\pi t)}{n\pi} \right]_0^1 + \left[\frac{\sin(n\pi t)}{n\pi} \right]_1^2$$

$$= \left[\frac{t \sin(n\pi t)}{n\pi} + \frac{\cos(n\pi t)}{n^2 \pi^2} \right]_0^1 + \left[\frac{\sin(n\pi t)}{n\pi} \right]_1^2$$

$$= \left[\left(\frac{\sin(n\pi)}{n\pi} + \frac{\cos(n\pi)}{n^2 \pi^2} \right) - \left(\frac{1}{n^2 \pi^2} \right) \right] + \left[\frac{\sin(2n\pi)}{n\pi} - \frac{\sin(n\pi)}{n\pi} \right]$$

$$0 + \frac{(-1)^n}{n^2 \pi^2} - \frac{1}{n^2 \pi^2} + 0 - 0$$

$$= \frac{-1}{n^2 \pi^2} = \frac{(-1)^n}{n^2 \pi^2}$$

$(-1)^n$ odd
even

$n = 1$

$$\frac{-t^2}{n^2 \pi^2}$$

$n = 2$

$$\frac{1}{n^2 \pi^2} = 0$$

$$b_n = \frac{2}{\pi} \left[\int_0^1 t \cdot \sin(n \cdot \pi \cdot t) + \int_1^2 1 \cdot \sin(n \cdot \pi \cdot t) \right]$$

$$= \left[t \cdot \frac{-\cos(n \pi t)}{n \pi} - \int \frac{1 \cdot -\cos(n \pi t)}{n \pi} \right]_0^1 + \left[\frac{\sin(n \pi t)}{n \pi} - \frac{\cos(n \pi t)}{n \pi} \right]_1^2$$

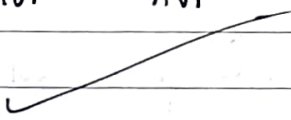
$$= \left[\frac{-t \cos(n \pi t)}{n \pi} + \frac{\sin(n \pi t)}{n^2 \pi^2} \right]_0^1 - \left[\frac{+\cos(n \pi t)}{n \pi} \right]_1^2$$

$$= \left[\left(\frac{-\cos(n \pi)}{n \pi} + \frac{\sin(n \pi)}{n^2 \pi^2} \right) - (0 + 0) \right] - \left[\frac{\cos(2n \pi)}{n \pi} - \frac{\cos(n \pi)}{n \pi} \right]$$

$$= \left[0 - \frac{(-1)^n}{n \pi} \right] - \left[\frac{1}{n \pi} - \frac{(-1)^n}{n \pi} \right]$$

$$= \frac{-(-1)^n}{n \pi} - \frac{1}{n \pi} + \frac{(-1)^n}{n \pi}$$

$$= \boxed{\frac{-1}{n \pi}}$$



~~4~~

Final fourier series representation is given by

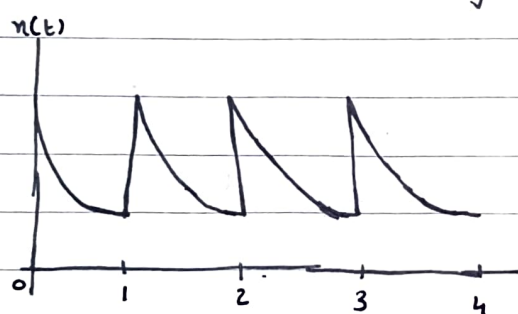
$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cdot \cos(n \omega t) + b_n \sin(n \omega t))$$

Exponential FS :: $\sum_{n=-\infty}^{\infty} C_n \cdot e^{jn\omega t}$

normal $x(t) = e^t$
decay $x(t) = e^{-t}$

$$\text{when } C_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega t} dt$$

Solve exponential FS for following wave form.



$$T = 1$$

$$\omega = 2 \cdot \pi \cdot f = 2 \cdot \pi \cdot \frac{1}{1} = 2\pi$$

$$x(t) = e^t \quad 0 \leq t \leq 1$$

$$\begin{aligned} C_n &= \frac{1}{1} \int_0^1 x(t) e^{-jn\omega t} dt \\ &= \int_0^1 e^t \cdot e^{-jn2\pi t} dt \\ &= \left[\frac{e^{t - jn2\pi t}}{-jn2\pi} \right]_0^1 \\ &= \frac{e^{1 - jn2\pi} - 1}{-jn2\pi} \\ &= \frac{1}{-jn2\pi} \left[e^{-jn2\pi} - 1 \right] \\ &= \frac{1}{-jn2\pi} \left[\cos(n2\pi) - j \sin(n2\pi) - 1 \right] \\ &= \frac{1}{-jn2\pi} \cdot [e^{-1} - 1] \Rightarrow \frac{1}{1 + jn2\pi} [e^{-1} - 1] \end{aligned}$$

Fourier Transform :

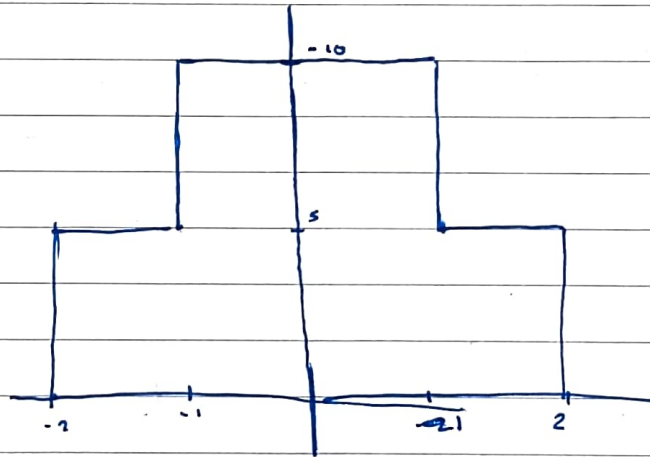
Convert from time domain to frequency domain.

$$F[x(t)] = X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

L, complex exponential term.

$$\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}, \quad \cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

Q1)



$$\int_{-2}^{-1} 5 \cdot e^{-j\omega t} dt + \int_{-1}^1 10 \cdot e^{-j\omega t} dt + \int_1^2 5 \cdot e^{-j\omega t} dt$$

$$= \frac{-5}{j\omega} \left[e^{-j\omega t} \right]_{-2}^{-1} + \frac{10}{-j\omega} \left[e^{-j\omega t} \right]_{-1}^1 + \frac{-5}{j\omega} \left[e^{-j\omega t} \right]_1^2$$

$$= \frac{-5}{j\omega} \left[e^{j\omega} - e^{2j\omega} \right] - \frac{10}{j\omega} \left[e^{-j\omega} - e^{j\omega} \right] - \frac{5}{j\omega} \left[e^{-2j\omega} - e^{-j\omega} \right]$$

$$= \frac{1}{j\omega} \left[-5e^{j\omega} + 5e^{2j\omega} - 10e^{-j\omega} + 10e^{j\omega} - 5e^{-2j\omega} + 5e^{-j\omega} \right]$$

$$= \frac{1}{j\omega} \left[(5e^{j\omega} - 5e^{-j\omega}) + (5e^{2j\omega} - 5e^{-2j\omega}) \right] \Rightarrow \frac{(5e^{j\omega} - 5e^{-j\omega})}{j\omega} + \frac{(5e^{2j\omega} - 5e^{-2j\omega})}{j\omega}$$

$$= \frac{10}{\omega} \sin(\omega) + \frac{10}{\omega} \sin(2\omega) = \frac{10}{\omega} \sin(\omega)$$

$$= \frac{10}{\omega} (\sin(\omega) + \sin(2\omega))$$

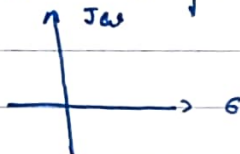
Laplace Transform:

$$L[x(t)] = X(s) = \int_{-\infty}^{\infty} x(t) \cdot e^{-st} dt$$

$X(s)$ is laplace transform of x of t

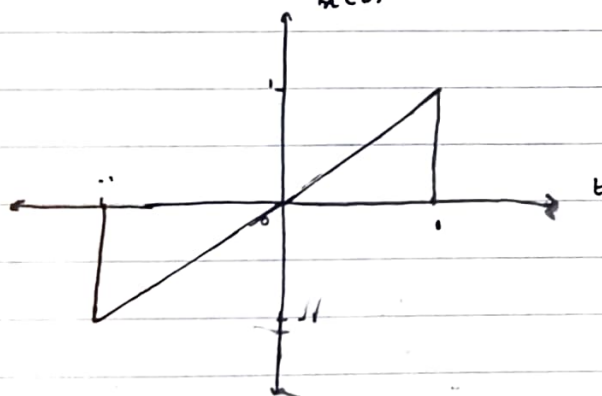
s laplace transform coefficient. $s = (\sigma + j\omega)$

s is in x - y plane, $x = \sigma$, $y = j\omega$



$$\sin \theta = \frac{e^{\theta} - e^{-\theta}}{2}, \quad \cos \theta = \frac{e^{\theta} + e^{-\theta}}{2}$$

Q1> Find laplace transform for given signal $x(t)$



$$x(t) = \begin{cases} -1 < t < 1 \\ 1 & 1 < t < 2 \end{cases}$$

$$= \int_{-\infty}^{\infty} x(t) \cdot e^{-st} dt$$

$$= \left[\frac{t \cdot e^{-st}}{-s} - \int \frac{e^{-st}}{-s} \right]_{-1}^1 \Rightarrow \left[\frac{-t e^{-st}}{s} + \frac{1}{s} \left(\frac{e^{-st}}{-s} \right) \right]$$

$$= \left[\frac{-t e^{-st}}{s} - \frac{e^{-st}}{s^2} \right]_{-1}^1$$

$$= \left[\frac{-e^{-s}}{s} - \frac{e^{-s}}{s^2} - \left(\frac{-e^s}{s} - \frac{e^s}{s^2} \right) \right] = \left(\frac{e^s}{s} - \frac{e^{-s}}{s^2} \right) - \left(\frac{e^{-s}}{s} + \frac{e^{-s}}{s^2} \right)$$

$$= \frac{2 \sin(s)}{s^2} - \frac{2 \cos(s)}{s}$$

Laplace Transform of Basic Functions.

Formulas ::

$$L[k] = \frac{k}{s}$$

$$L[e^{at}] = \frac{1}{s-a}$$

$$L[e^{-at}] = \frac{1}{s+a}$$

$$L[\sin at] = \frac{a}{s^2 + a^2}$$

$$L[\cos at] = \frac{s}{s^2 + a^2}$$

$$L[\sinh at] = \frac{a}{s^2 - a^2}$$

$$L[\cosh at] = \frac{s}{s^2 - a^2}$$

Properties ::

① Linearity

$$L[ax_1(t) + bx_2(t)] = ax_1(s) + bx_2(s)$$

② Time Shift

$$L[x(t-k)] = e^{-sk} \cdot X(s)$$

③ Differentiation

$$L[x(t) \cdot t^n] = (-1)^n \frac{d^n}{ds^n} X(s)$$

④ Scaling

$$L[x(at)] = \frac{1}{a} \cdot X\left(\frac{s}{a}\right)$$

⑤ First Shift

$$L[x(t) \cdot e^{at}] = X(s-a)$$

⑥ Convolution

$$L[x(t) * h(t)] = X(s) \cdot H(s)$$

⑦ Division by 't' property

$$L \left[\frac{x(t)}{t} \right] = \int_s^{\infty} X(s) ds$$

⑧ Integral Property

$$\mathcal{L} \left[\int_0^t x(t) \cdot dt \right] = \frac{1}{s} \cdot X(s)$$

Q1) Find $L \left[\int_0^t e^{-3t} \sin(4t) \cdot t \, dt \right]$

$$= \textcircled{1} : \mathcal{L}[\sin 4t] = \frac{4}{s^2 + 16}$$

$$\textcircled{b} = \mathcal{L} \left[\int_0^t e^{-3t} \cdot \frac{4}{s^2+16} t' dt \right] \quad \text{Differentiation}$$

$$= \frac{8s}{(s^2 + 16)^2}$$

③: $\int_{-\infty}^{\infty} e^{-st} \cdot \frac{8s}{(s^2+16)^2} dt$

$$= \frac{8(s+3)}{((s+3)^2 + 16)^2} = \frac{8s+24}{((s+3)^2 + 16)^2} \cdot \frac{8s+24}{(s^2 + 6s + 25)^2}$$

$$(4) \quad L \left[\begin{matrix} 1 \\ 0 \end{matrix} \right] \frac{8s+24}{(s^2+6s+25)^2} \right]$$

$$\frac{1}{s} \cdot \frac{8s + 24}{(s^2 + 6s + 25)^2}$$

$$\frac{8s+24}{s(s^2+6s+25)^2}$$

Z-Transform For DTS

$$X[n] = \underset{x[1]}{1}, \underset{x[2]}{2}, \underset{x[3]}{10}, \underset{x[4]}{11} \quad \text{Find } Z \text{ transform.}$$

$$Z[X[n]] = X(Z) = \sum_{n=-\infty}^{\infty} x[n] \cdot Z^{-n}$$

$$= \sum_{n=0}^3 : x[0] \cdot Z^0 + x[1] \cdot Z^{-1} + x[2] \cdot Z^{-2} + x[3] \cdot Z^{-3}$$

$$= 1 + 2 \cdot Z^{-1} + 10 Z^{-2} + 11 Z^{-3}$$

$$= \frac{1Z^3}{Z^3} + \frac{2Z^2}{Z^3} + \frac{10Z}{Z^3} + \frac{11}{Z^3}$$

$$= \frac{Z^3 + 2Z^2 + 10Z + 11}{Z^3}$$

$$X[n] = \underset{x[-1]}{4}, \underset{x[0]}{5}, \underset{x[1]}{12}, \underset{x[2]}{21}$$

$$= \sum_{n=-2}^2 : x[-2] \cdot Z^2 + x[-1] \cdot Z^1 + x[0] \cdot 1 + x[1] \cdot Z^{-1}$$

$$= 4Z^2 + 5Z + 12 + 21Z^{-1}$$

$$= \frac{4Z^3}{Z} + \frac{5Z^2}{Z} + \frac{12Z}{Z} + \frac{21}{Z}$$

$$= \frac{4Z^3 + 5Z^2 + 12Z + 21}{Z}$$